

1. Administrative & Study Identification

Full Study Title: Observational Study of Nivolumab Monotherapy or in Combination With Ipilimumab in Participants With Advanced Melanoma and in Participants With Adjuvant Nivolumab Therapy (NICO Study) **Short Title/Acronym:** NICO Study / CA209-654 **Protocol Number:** CA209-654 **NCT Number:** NCT02990611 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab (anti-PD1) alone or in combination with Ipilimumab (anti-CTLA4) **Phase of Development:** Observational / Phase 4 Equivalent **Medical Monitor:** Bristol Myers Squibb Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world overall survival (OS) in patients with advanced melanoma receiving nivolumab in combination with ipilimumab. **Secondary Objectives:** To evaluate overall survival (OS) in patients treated with nivolumab monotherapy, progression-free survival (PFS), objective response rates (ORRs), safety profiles (including immune-related adverse events), and health-related quality of life (HRQoL).

2.2 Background & Rationale While the combination of nivolumab and ipilimumab or nivolumab monotherapy has shown robust efficacy in randomized controlled trials for advanced melanoma, real-world data covering diverse patient subgroups—such as those with or without symptomatic or asymptomatic melanoma brain metastases (MBM)—are limited. The prospective NICO study evaluates real-world effectiveness, safety, and patient-reported outcomes to validate the long-term clinical benefit of these regimens in an unselected, routine clinical practice population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational, Prospective, Non-interventional **Control Type:** Active Comparator (Real-world cohort comparison: NIVO+IPI vs. NIVO alone) **Blinding:** Open-label **Randomization:** N/A (Observational routine clinical practice)

3.2 Planned Enrollment & Duration Target \$N\$: 755 patients (n = 486 receiving nivolumab + ipilimumab; n = 269 receiving nivolumab alone) **Number of Sites:** Multicenter (Germany) **Estimated Study Start:** 2016 **Estimated Primary Completion:** July 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (≥ 18 years of age). 2. Confirmed diagnosis of advanced (unresectable or metastatic) melanoma, including participants with adjuvant nivolumab therapy. 3. Initiating treatment with nivolumab monotherapy or nivolumab in combination with ipilimumab (any-line) per routine clinical practice. 4. Patients with or without melanoma brain metastasis (MBM) are eligible (both asymptomatic and symptomatic).

4.2 Exclusion Criteria 1. Participation in another interventional therapeutic clinical trial. 2. Any condition that, according to the treating physician, would contraindicate the use of nivolumab or ipilimumab based on the approved local product labeling.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab monotherapy or Nivolumab combined with Ipilimumab administered at approved doses according to standard of care/physician discretion. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or standard completion of adjuvant therapy as per routine clinical practice.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, including corticosteroids (e.g., dexamethasone) for the management of symptomatic brain metastases or immune-related adverse events (irAEs). **Prohibited:** Concurrent investigational anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Baseline	Day 0	Treatment initiation, baseline characteristic assessment, ECOG PS, MBM status documentation.
Follow-Up Visits	Per Routine Care	Safety and effectiveness assessments (e.g., tumor imaging, overall survival tracking), HRQoL questionnaires.
End of Study	Up to 5 years	Final survival status, long-term follow-up, tracking of late-onset toxicities.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS) in patients receiving nivolumab in combination with ipilimumab, defined as the time from the initiation of treatment to death from any cause. **Measurement Method:** Clinical records and survival tracking in routine clinical practice.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) in patients receiving nivolumab alone. 2. Objective Response Rate (ORR) and Progression-Free Survival (PFS) for both treatment cohorts. 3. Incidence and severity of treatment-related adverse events (TRAEs), specifically immune-related AEs (e.g., colitis, endocrinopathies, hepatitis, pneumonitis). 4. Health-Related Quality of Life (HRQoL) outcomes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Real-world tracking of immune-related adverse events (irAEs) via treating physician assessments. Data collected on specific preferred terms including colitis, endocrinopathies, hepatitis, nephritis/renal dysfunction, pneumonitis, and rash. **Serious Adverse Events (SAEs):** Grade 3/4 treatment-related adverse events reported and tracked longitudinally. **Data Monitoring Committee (DMC):** Safety profiles and adverse event management assessed via standard pharmacovigilance mechanisms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-treat equivalent encompassing all enrolled patients who received at least one dose of the respective immunotherapies. **Sample Size Justification:** The enrolled real-world cohort of 755 patients provided sufficient observational power to evaluate long-term outcomes and safety metrics (including landmark 3-year OS rates) across distinct subgroups. **Handling of Missing Data:** Handled per standard statistical protocols for observational registry data, utilizing Kaplan-Meier methods for survival estimations and multivariable Cox regression models.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and applicable non-interventional/observational study regulations. **Monitoring Plan:** Data collected prospectively from electronic health records and routine clinical visits across participating German centers. **Audit Trail:** Secure database lock and routine quality audits in alignment with Bristol Myers Squibb registry standards.

11. Study Identifiers & Governance

Primary ID: CA209-654 **Registry Number:** NCT02990611 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** NICO Study (Real-

Life Efficacy and Safety of Nivolumab Monotherapy or in Combination With Ipilimumab in Patients With Advanced Melanoma)
Official Regulatory Title: Observational Study of Nivolumab Monotherapy or in Combination With Ipilimumab in Participants With Advanced Melanoma and in Participants With Adjuvant Nivolumab Therapy

12. Clinical Framework

Primary Condition: Advanced (Unresectable or Metastatic) Melanoma
Diagnosis Threshold: Histologically confirmed advanced melanoma requiring systemic treatment. **Medical Methodology:** Immune Checkpoint Inhibition (anti-PD-1 +/- anti-CTLA-4). **Optimization Target:** Maximizing overall survival (OS) and intracranial/extracranial tumor response while managing immune-related adverse events.

13. Study Procedures & Methodology

Management Pathway: Standard routine oncological clinical pathway for advanced melanoma. **Education Protocol (HCP):** Standard guidelines for the management of immune-related adverse events associated with immune checkpoint inhibitors. **Education Protocol (Patient):** Routine counseling regarding the risks and benefits of immunotherapy. **Quality Audit Mechanism:** Registry data quality control and ongoing outcome tracking up to 5 years.

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years of longitudinal follow-up. **Onsite Visit Cadence:** As per routine clinical standard of care for patients on active immunotherapy and subsequent surveillance. **Remote Monitoring Frequency:** N/A (Observational electronic data capture). **Checklist Review Interval:** Data cut-offs established for interim and final analyses.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				